

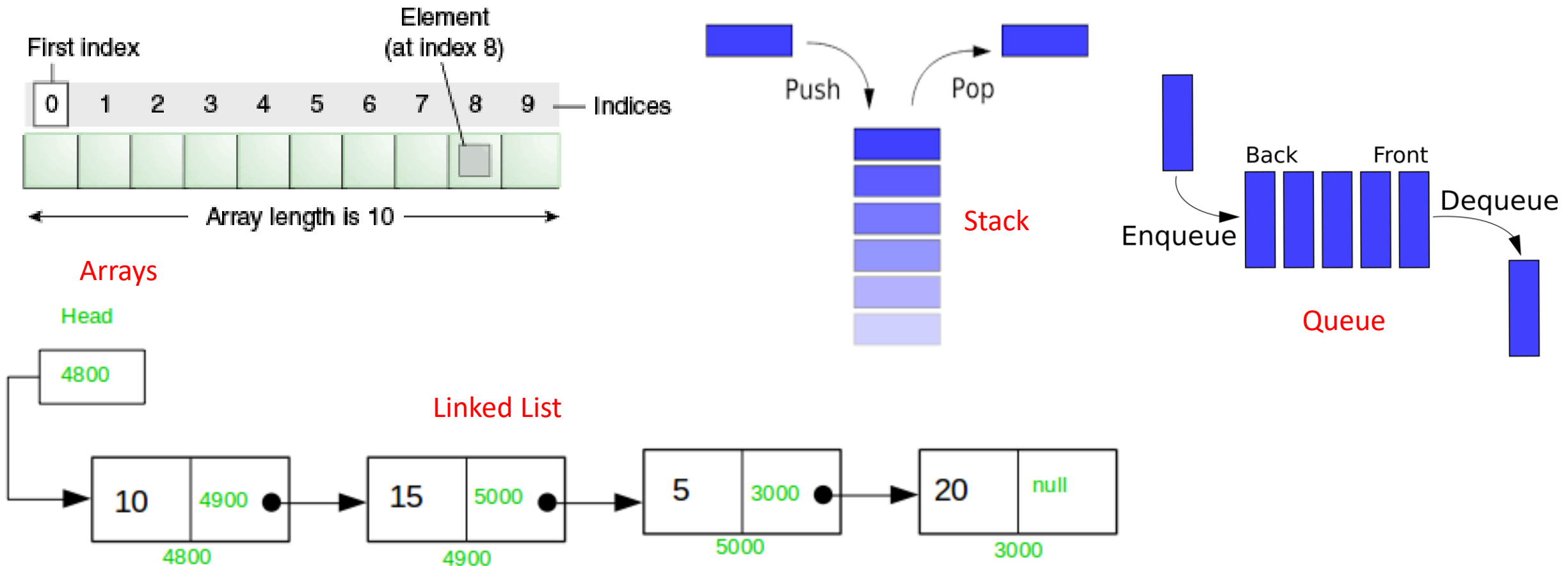


Introduction to trees

MODULE 4 PART I

Introduction to Trees

Linear Data Structures

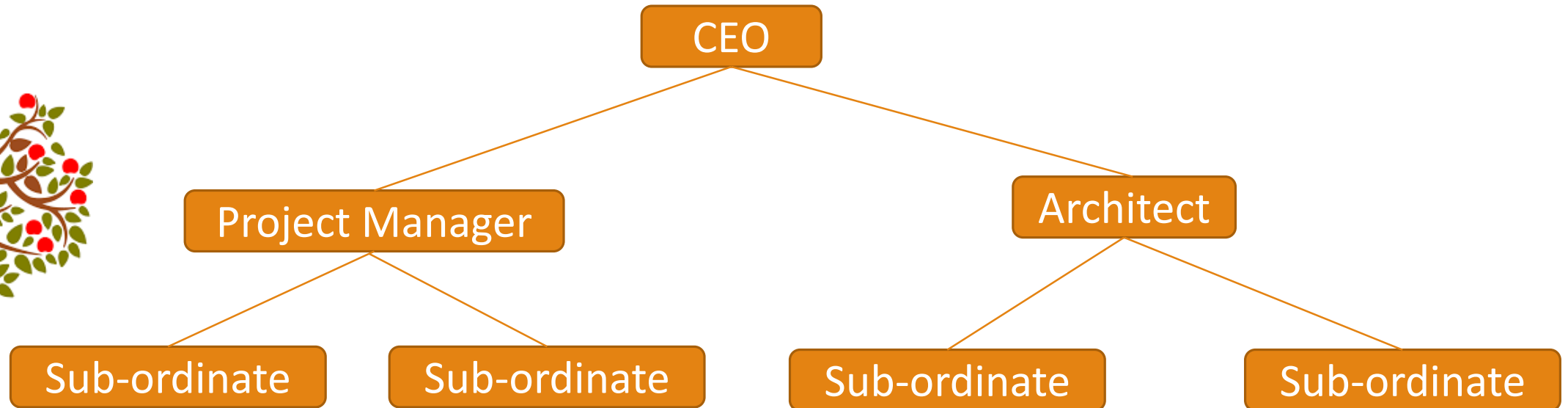


The Choice of data structures depends on..

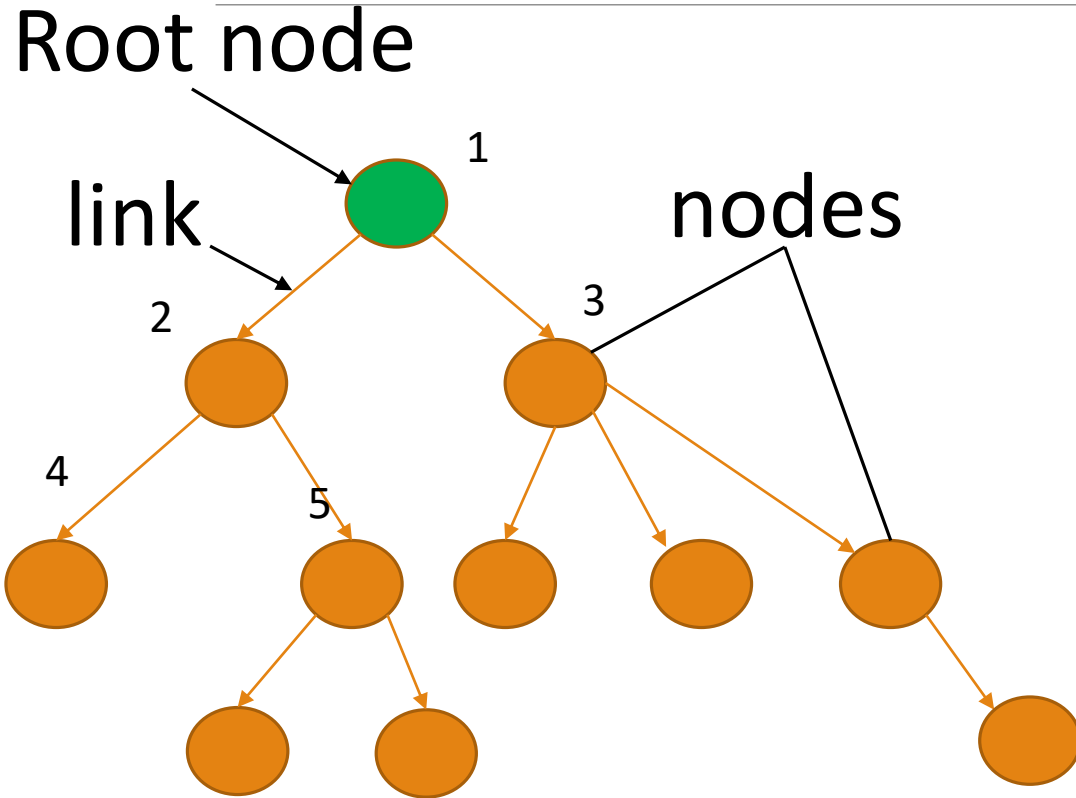
- What need to be stored.
- Cost of operations
- Memory usage
- Ease of Implementation

Trees

➤ To represent hierarchical data.



Trees cntd..



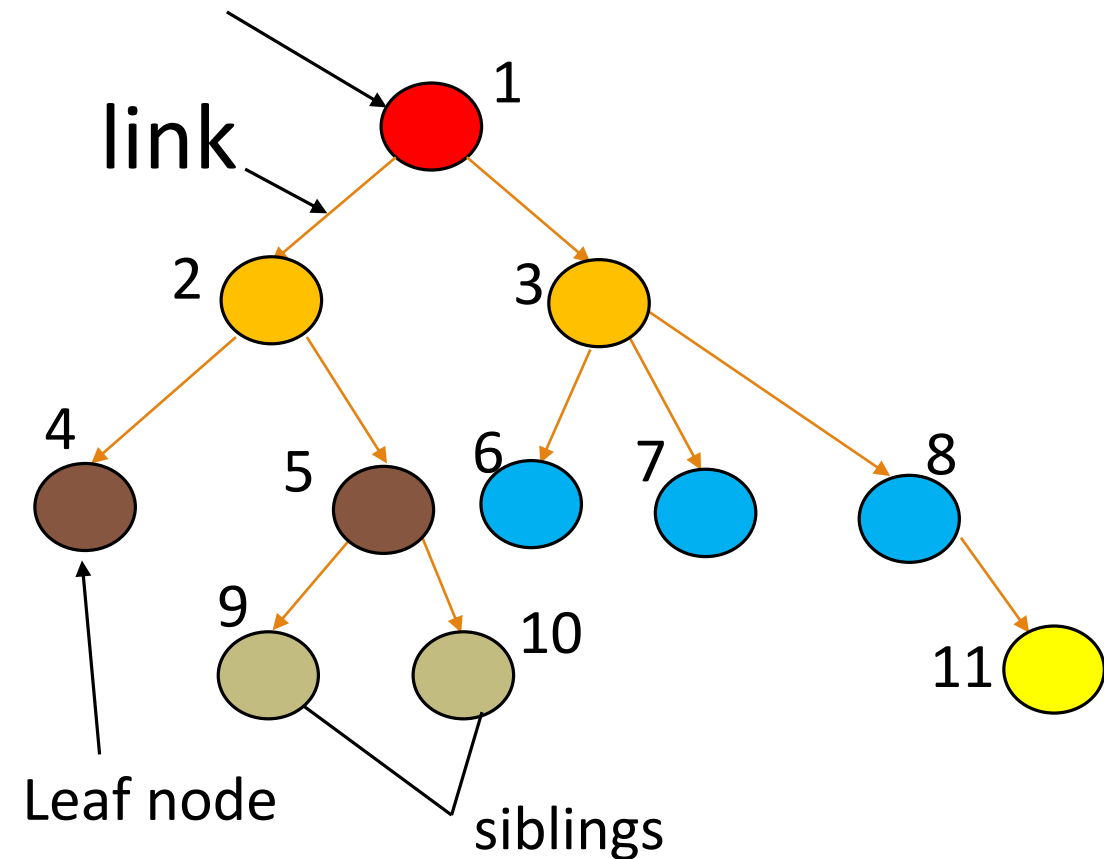
- Collection of entities (nodes) linked together to represent a hierarchy.
- It's a hierarchical structure
- The top node in the tree is called root node
- Each node consists of different kinds of data
- Root node has children's

Tree terminology

- root: only node without a parent
- children
- parent

Root node

link



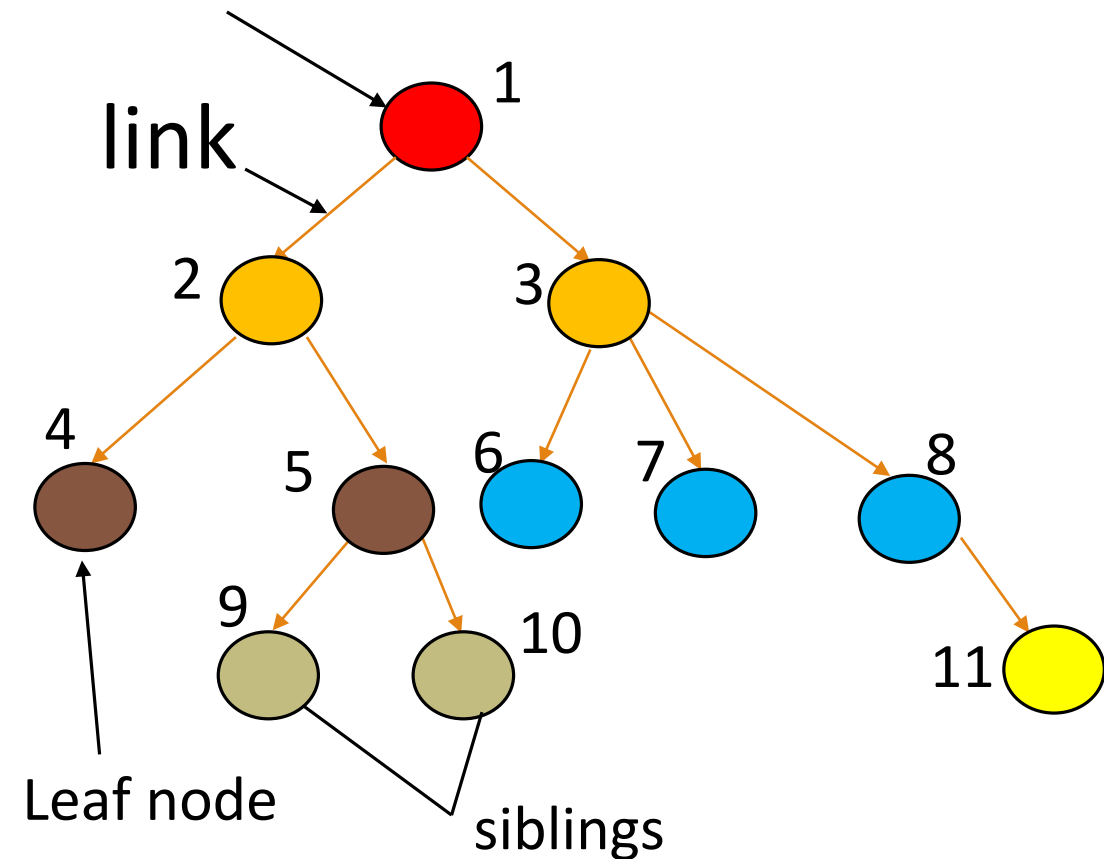
- sibling: children of same parent
- parent-child relationship: If a node have a direct link to some other node
- leaf: Any node in the tree does not have a child
- Internal node: All other nodes with at-least one child
- Parent of parent is grand parent
- Ancestor and Descendants

Tree terminology

Root node

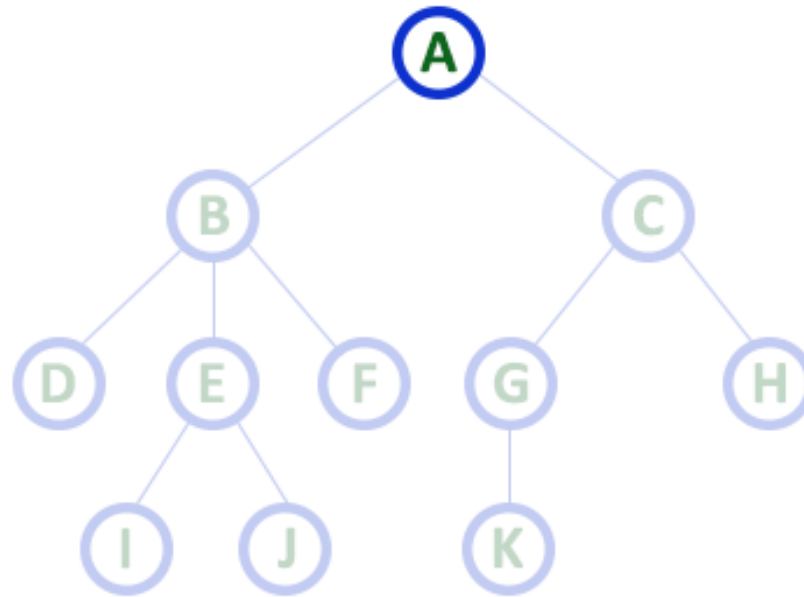
link

- Ancestor and Descendants
- Cousins: nodes having same grand parent



Root

- In a tree data structure, the first node is called as Root Node.
 - Every tree must have root node.
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- root node is the origin of tree data structure.
 - In any tree, there must be only one root node.

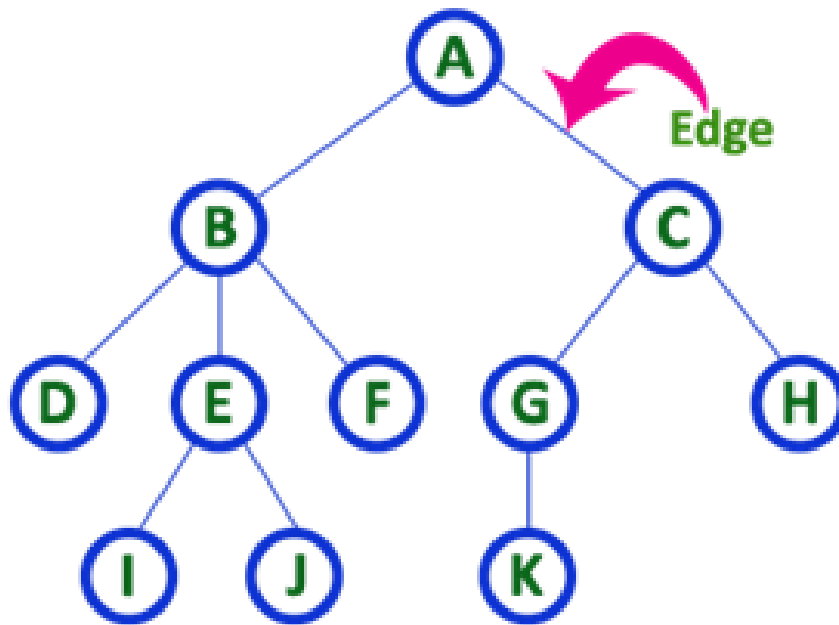


Here 'A' is the 'root' node

- In any tree the first node is called as ROOT node

Edge

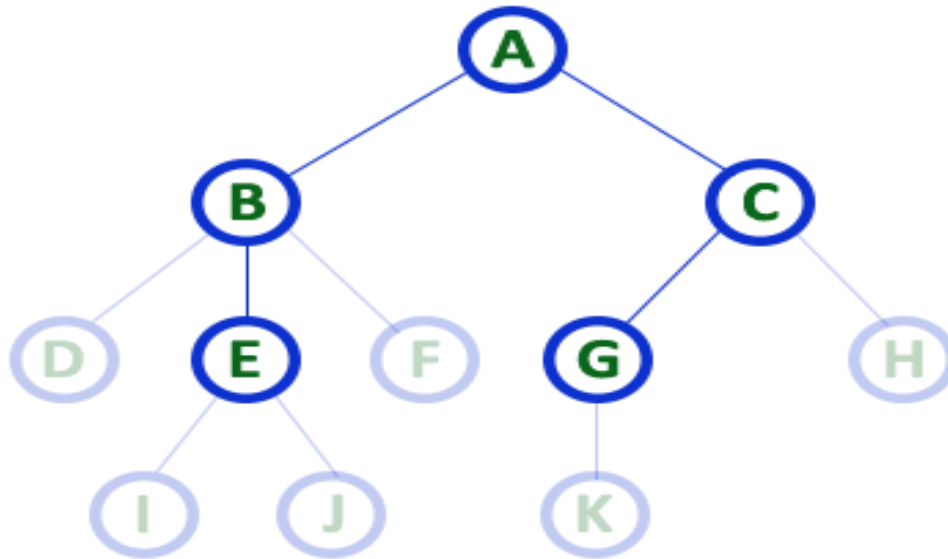
- In a tree data structure, the direct connecting link between any two nodes is called as EDGE.
 - In a tree with 'N' number of nodes there will be a maximum of 'N-1' number of edges.
-



- In any tree, 'Edge' is a connecting link between two nodes.

Parent

- In a tree data structure, the node which is predecessor of any node is called as PARENT NODE.
 - In simple words, the node which has branch from it to any other node is called as parent node.
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- Parent node can also be defined as "The node which has child / children".

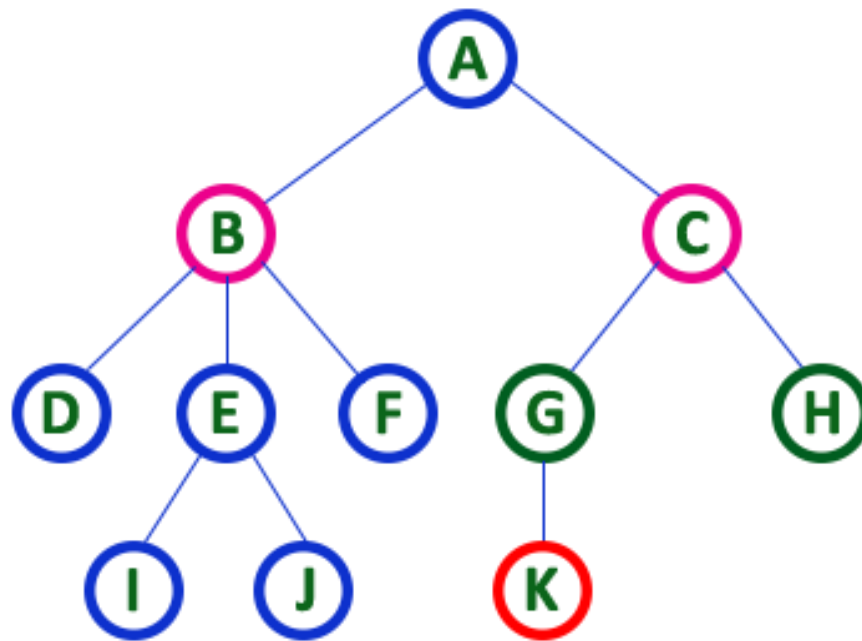


Here A, B, C, E & G are **Parent** nodes

- In any tree the node which has child / children is called '**Parent**'
- A node which is predecessor of any other node is called '**Parent**'

Child

- In a tree data structure, the node which is descendant of any node is called as CHILD Node.
 - In simple words, the node which has a link from its parent node is called as child node.
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- In a tree, any parent node can have any number of child nodes. In a tree, all the nodes except root are child nodes.



Here **B & C** are **Children of A**

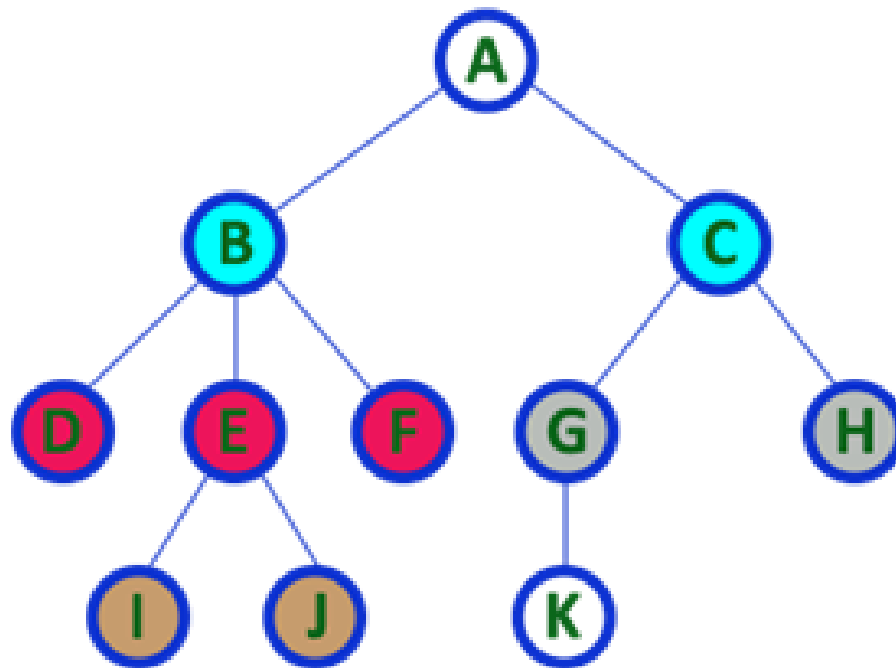
Here **G & H** are **Children of C**

Here **K** is **Child of G**

- descendant of any node is called as **CHILD Node**

Siblings

- In a tree data structure, nodes which belong to same Parent are called as SIBLINGS.
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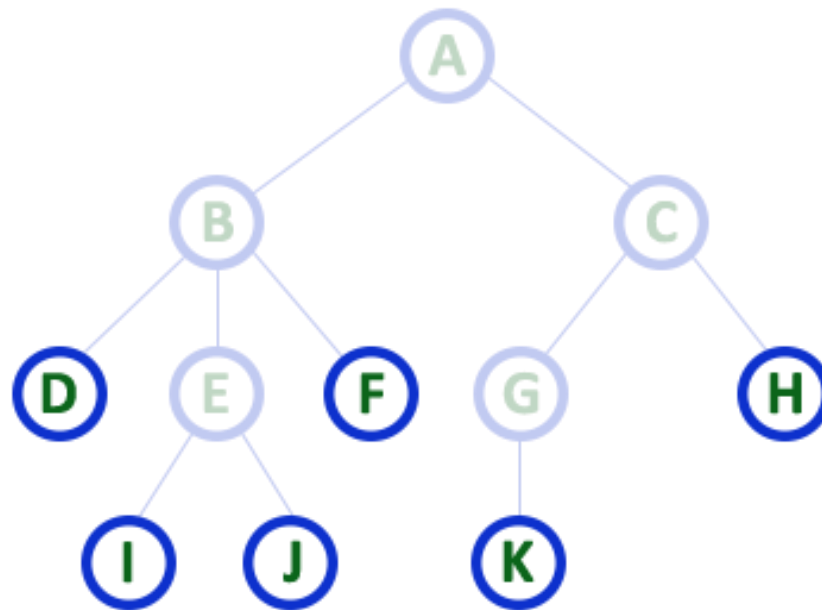
Here **B & C** are **Siblings**
Here **D E & F** are **Siblings**
Here **G & H** are **Siblings**
Here **I & J** are **Siblings**

- In any tree the nodes which has same Parent are called '**Siblings**'
- The children of a Parent are called '**Siblings**'

Leaf

- In a tree data structure, the node which does not have a child is called as LEAF. the leaf nodes are also called as External Nodes.
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- External nodes is also a node with no child. In a tree, leaf node is also called as 'Terminal' node.

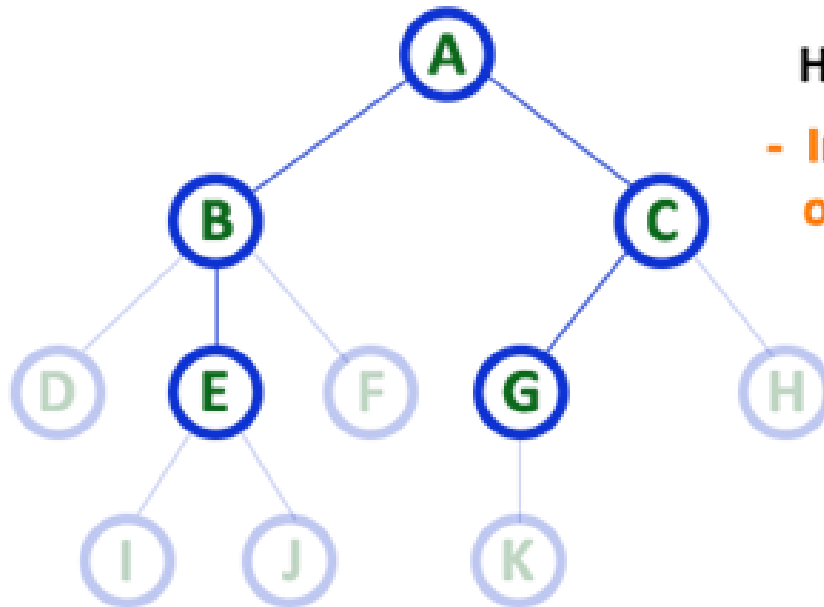


Here D, I, J, F, K & H are **Leaf** nodes

- In any tree the node which does not have children is called '**Leaf**'
- A node without successors is called a '**leaf**' node

Internal nodes

- In a tree data structure, the node which has at least one child is called as INTERNAL Node.
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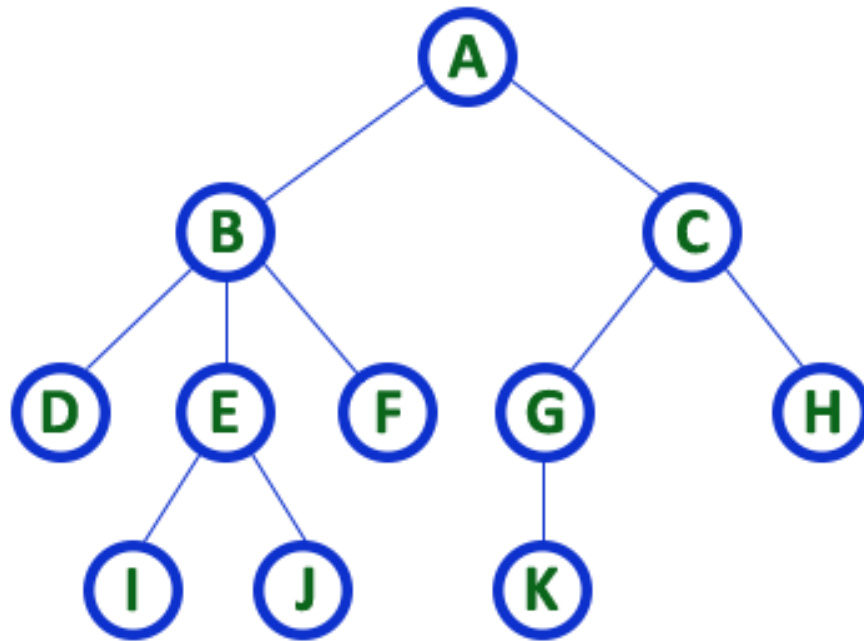
Here A, B, C, E & G are **Internal** nodes

- In any tree the node which has atleast one child is called '**Internal**' node
- Every non-leaf node is called as '**Internal**' node

Degree

- In a tree data structure, the total number of children of a node is called as DEGREE of that Node.
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- The highest degree of a node among all the nodes in a tree is called as 'Degree of Tree'.



Here **Degree** of B is 3

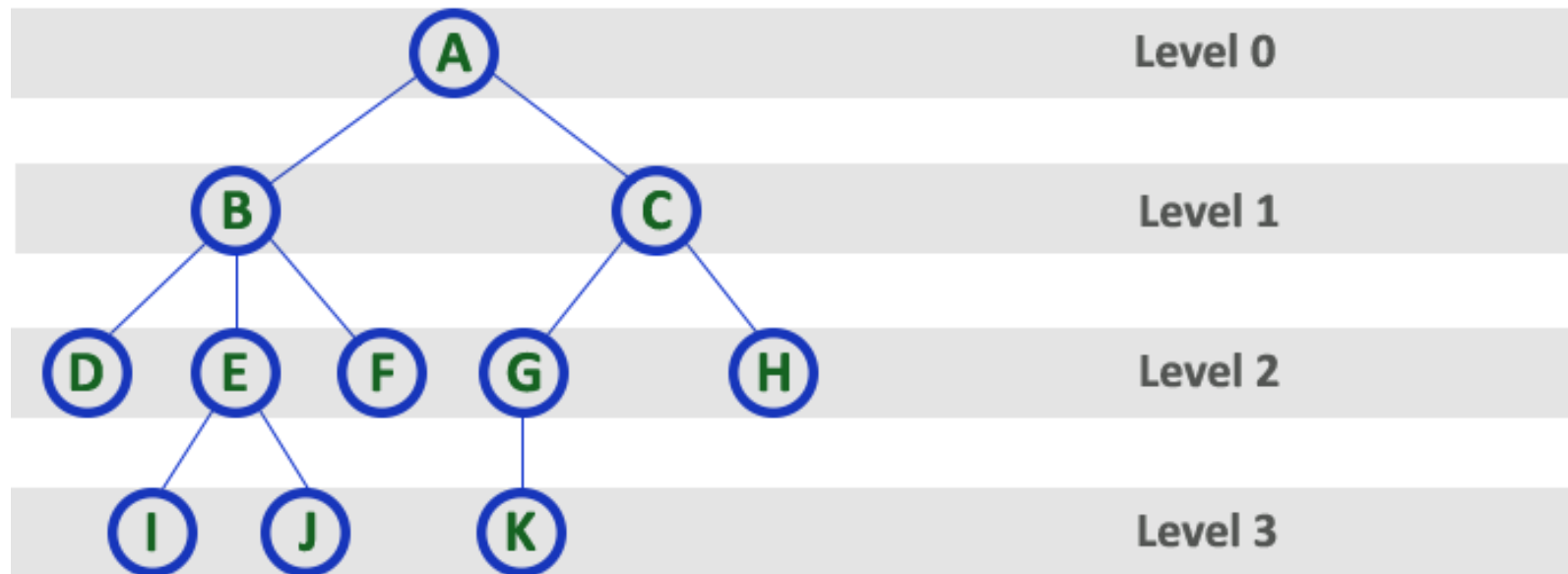
Here **Degree** of A is 2

Here **Degree** of F is 0

- In any tree, '**Degree**' a node is total number of children it has.

Level

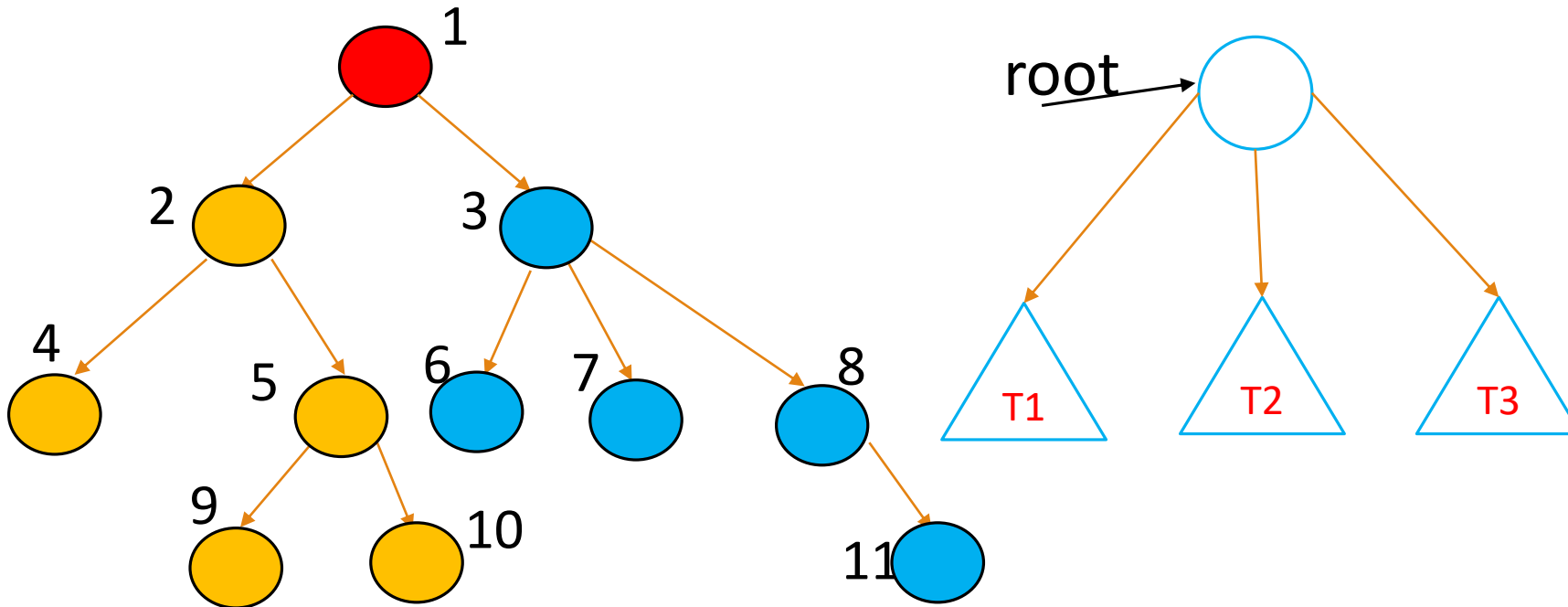
- In a tree data structure, the root node is said to be at Level 0 and the children of root node are at Level 1 and the children of the nodes which are at Level 1 will be at Level 2 and so on... In simple words, in a tree each step from top to bottom is called as a Level and the Level count starts with '0' and incremented by one at each level (Step).



Tree terminology cntd..

➤ Tree is a recursive data structures

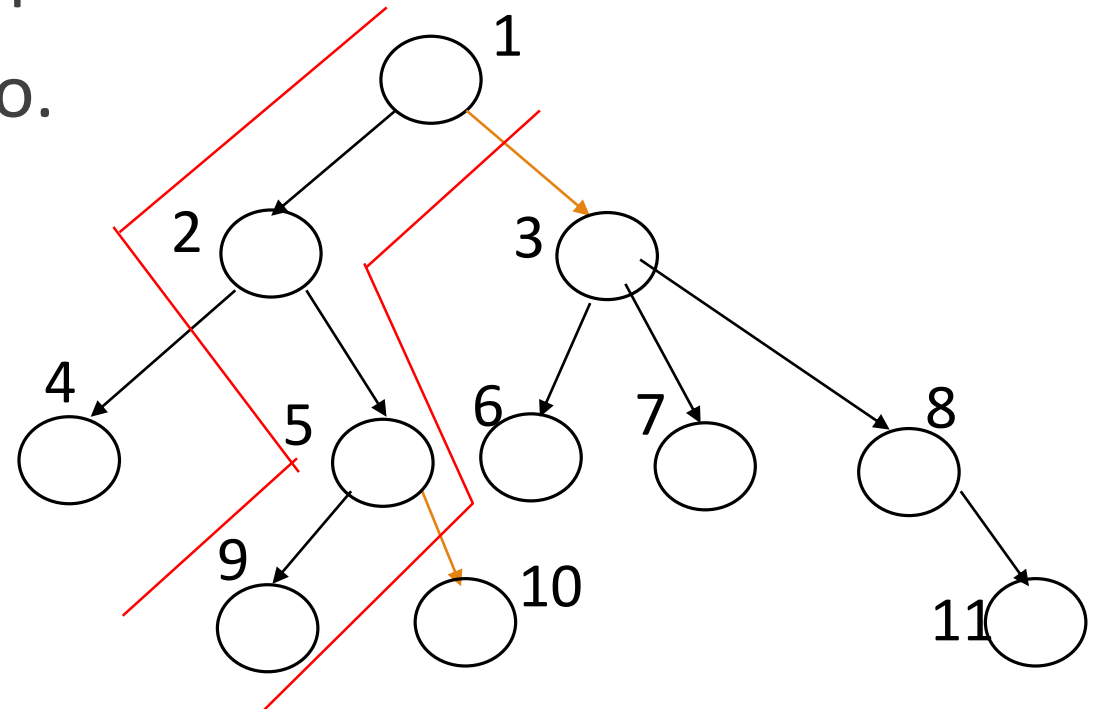
➤ A tree is a recursive data structure which consists of a distinguished node called root and some sub-trees in such a way that root of the tree contains link to roots of all the sub-trees.



Tree terminology cntd..

- **Depth** of a node (x) is the length of the path from root to x.
- **Length** is number of edges in path from root to x.
- The depth of root node is zero.

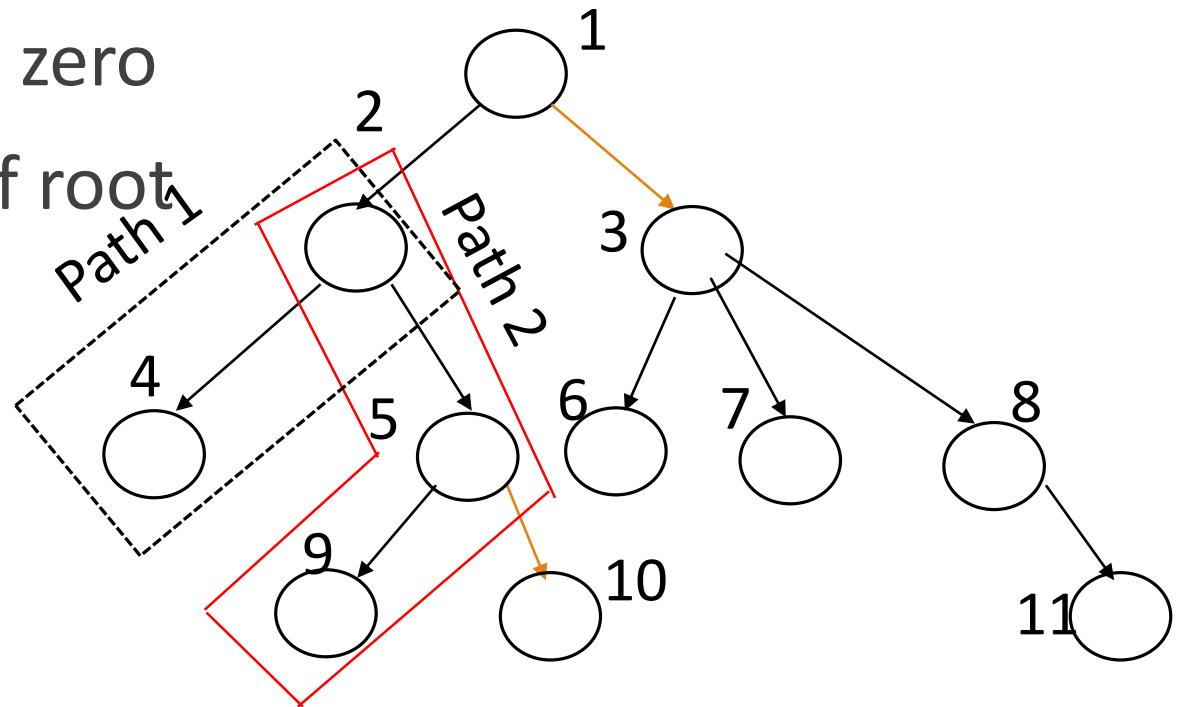
Root 1 – 2 – 4



Tree terminology cntd..

- **Height** of a node (x) is the number of edges in **longest path** from x to a leaf.
- Height of leaf nodes will be zero
- Height of a tree is height of root node

Maximum dept of a tree is
Equals to height of the tree



Sub tree

In a tree data structure, each child from a node forms a subtree recursively.

